

Sensitivity Comparison of Atomic Interferometers for Gravitational Measurements on Positronium or Treatise on Shelling Bananas

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what is positronium

why is it interesting

how to do it

why is it hard



problem



figure: a banana



problem

type	result	probability
β^-	^{40}Ca	89.6 %
EC	$^{40}\text{Ar}^*$	10.3 %
direct EC	^{40}Ar	0.1%
β^+	^{40}Ar	0.001%



figure: a banana, and ^{40}K decay modes table



what is positronium?

$\underbrace{\text{positr}}_{e^+}$ $\underbrace{\text{onium}}_{e^-}$ bound state

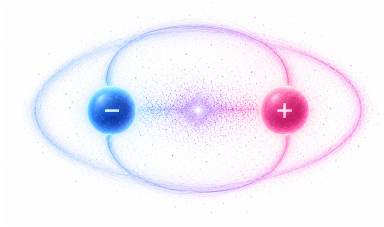


figure: artistic image of positronium



positronium states

and a quick repeat from atomic physics

- ▶ para-positronium 1^1S_0 : $T_{1/2} = 125$ ps
- ▶ ortho-positronium 1^3S_1 : $T_{1/2} = 142$ ns



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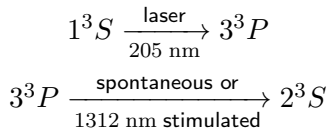
what does this mean?

$$n^{2S+1}L_J$$

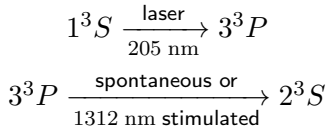
$$\vec{J} = \vec{L} + \vec{S}$$



how to obtain meta-stable beam?



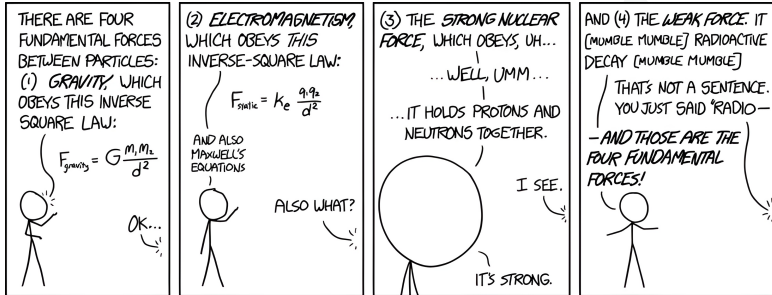
how to obtain meta-stable beam?



- ▶ Doppler velocity selection
- ▶ slit collimation



fundamental forces in physics



fundamental forces in physics

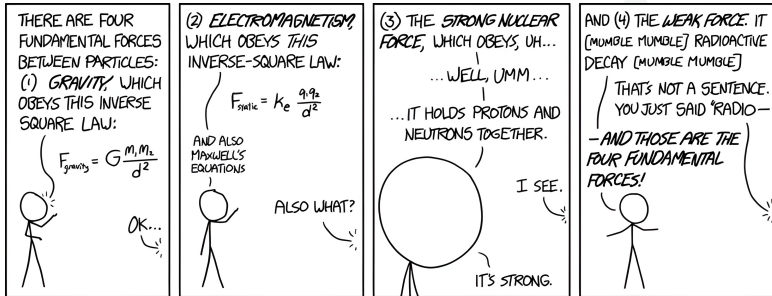


figure: "Of these four forces, there's one we don't really understand."
"Is it the weak force or the strong—" "It's gravity."



why do we bother?

*'But in any event, it is in the hands of our generation to perform an experiment to measure the gravitational acceleration of antimatter. Some day it will be done, whether we do it or not. **It will be done.** If we do not do it, and the answer eventually turns out to be what we expect, then future generations will look back upon us and say it was a shame. But if the answer turns out to be a surprise, then, if we do not do it, future generations will look back upon us and say we were fools.'*

- M. M. Nieto et al. Theoretical Motivation for Gravitation Experiments on UltraLow Energy Antiprotons and Antihydrogen. arXiv, 1994.



a bit of theory

why are we even bothering to measure it?

- ▶ **CPT symmetry:** according to the CPT theorem, antimatter should experience the same gravitational acceleration as matter
- ▶ **photon behaviour:** photons are their own antiparticles, and they act just like General Relativity laws tell them so
- ▶ **conservation of energy:** if antimatter goes up, we can annihilate it "higher" than it was produced



a bit of theory...

However...

- ▶ **Kowitt theory:** antiparticles are "holes" in Dirac particle sea with negative gravitational mass
- ▶ **Santilli-Villata theory:** antimatter exists in inverted space-time, which is projected to our reality as negative motion
- ▶ **Cabbolet's theory:** particles move in "stepwise" motion, alternating between state of rest and "wave-like" movement
- ▶ **dark matter???**



Moiré deflectometer

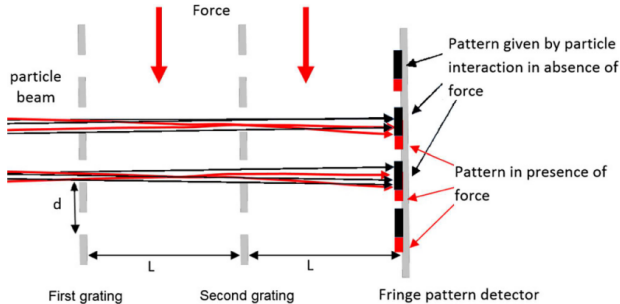
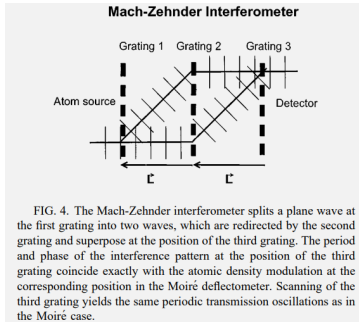


Fig. 3. Schematic of a classical two-grating deflectometry system with periodicity d .

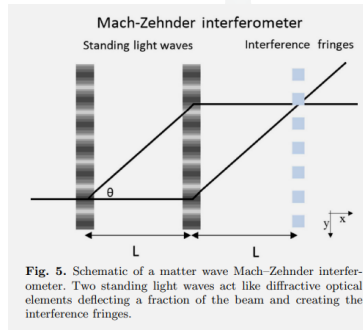
figure: (source: Mariazzi et al., 2020)



Mach-Zehnder interferometer



(a) (source: Oberthaler et al., 1996)



(b) (source: Mariazzi et al., 2020)



Talbot-Lau interferometer

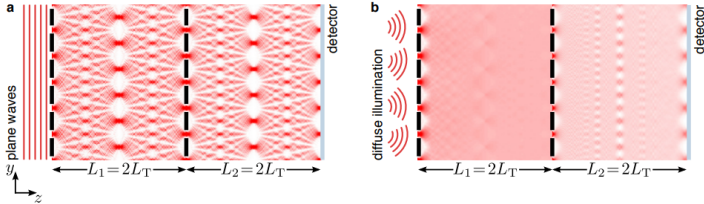


Figure 2.4.: (a) A Talbot-Lau interferometer requires a second grating. The distance between the two gratings and between the second grating and the observational plane equals an integer multiple of the Talbot length. For plane waves one observes the same Talbot carpet but in contrast to the simple Talbot effect, the fringe pattern observed on the detector remains when using a diffuse illumination as shown in (b). The reason is the spatial coherence generated by the first grating. For better perceptibility, the intensity of the light field between second grating and the detector is divided by the open fraction η to compensate for the intensity loss at the passage of the second grating.

figure: (source: Bräunig, 2015)



why is it hard

or: why hasn't it been done before

device	1 g measurement time
moiré	22000 yr
Mach-Zehnder	5.8 yr
Talbot-Lau	730 yr



bibliography



Mariazzi, Sebastiano et al. (Apr. 2020). "Toward inertial sensing with a 23S positronium beam". en. In: *The European Physical Journal D* 74.4, p. 79. ISSN: 1434-6060, 1434-6079. DOI: 10.1140/epjd/e2020-100585-8. URL: <https://link.springer.com/10.1140/epjd/e2020-100585-8>.



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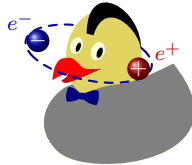


Bräunig, Philippe H. M. (2015). "Atom Optical Tools for Antimatter Experiments". In: DOI: 10.11588/HEIDOK.00018044. URL: <http://archiv.ub.uni-heidelberg.de/volltextserver/id/eprint/18044>.



the end

thank you for attention



<https://github.com/falcotinnunculus/przezrocza/blob/wladca/electricboogaloo/prezka.pdf>



(backup) why is it hard 2: electric boogaloo

or: why haven't I done it yet

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